E-Commerce Customer Behavior Analysis

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B.TECH CSE (3RD YEAR)

Technical Stack

Excel · Python (Jupyter)
PostgreSQL · Tableau

Dataset

700 rows · 12 columns
Kaggle source

Objective

Identify revenue patterns, high-value segments, category & geographic insights

BUSINESS CONTEXT

- *E-commerce companies generate large volumes of transactional data.*
- *However, raw data alone does not support strategic decision making.*
- *This project focuses on converting operational transactional data into actionable business insights.*

PROBLEM STATEMENT

Key Questions

- Identify high-revenue customers
- Analyze category-level performance
- Evaluate country-wise revenue contribution
- Understand revenue trends and seasonality

Goal

- Enable data-driven decision making for business strategy

METHODOLOGY — OVERVIEW

❑ Data Cleaning (SQL)

- ➢ COUNT(), duplicate detection, NULL checks
- ➢ Ensured correct data types

❑ Validation (Python)

- ➢ Revenue = Unit Price × Quantity
- ➢ IQR method used for outlier detection

❑ Feature Engineering (SQL)

- ➢ Created Revenue column for business performance tracking
- ➢ Aggregated revenue using SUM() and GROUP BY
- ➢ Prepared monthly trend data for growth analysis

❑ Exploratory Data Analysis & Visualization

- ○ Line charts
- ○ Bar charts
- ○ Pie charts

❖ Applied IQR method in Python for outlier detection

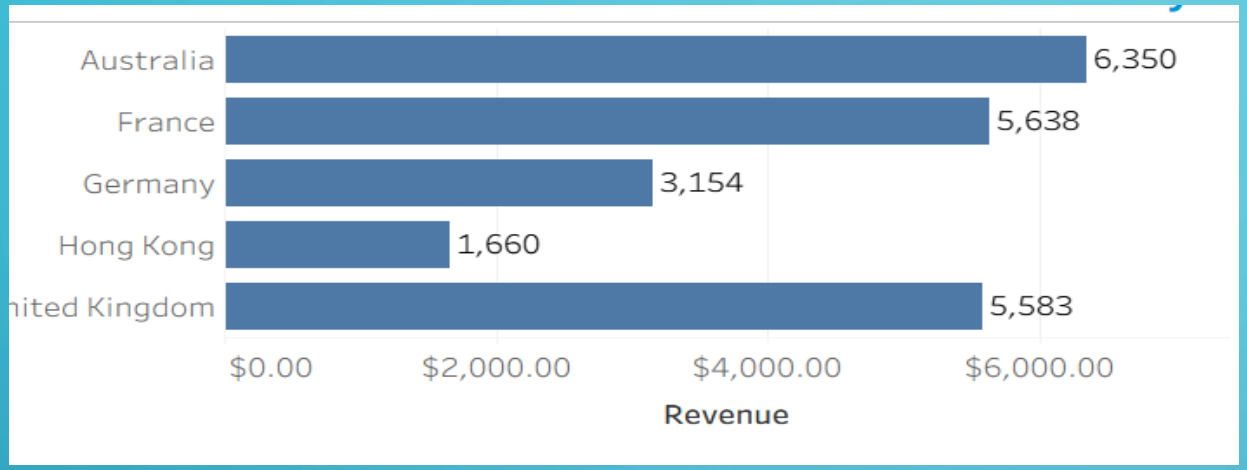
➞ Final insights visualized in Tableau Dashboard

KEY BUSINESS METRICS

❖ Key Observation

- ➢ Revenue peaks exceed $2.5K in top months with
- ➢ MoM growth spikes above 300% in some periods.

GEOGRAPHIC INSIGHTS



Country	Avg Order Val..	Total Quantity	Total Revenue
Australia	$226.78	5,325	$6,349.89
France	$40.27	3,623	$5,638.16
United Kingdom	$19.19	2,881	$5,582.93
Germany	$38.00	1,951	$3,153.71
Hong Kong	$237.18	826	$1,660.24

Australia

(~$6.3K) Top revenue contributor

Hong Kong

Highest AOV (~$237) indicating premium customer positioning

France & UK

Significant international contributors

INSIGHTS

Hong Kong represents a high-value premium market opportunity.

REVENUE DYNAMICS

- Volatility
- Revenue shows cyclical trends
- Peak months exceed $2.5K
- Growth Spikes
- Some months show >300% MoM growth
- Forecast Opportunity
- Demand forecasting can reduce revenue volatility

TECHNOLOGY STACK & IMPLEMENTATION

- Microsoft Excel
- SQL
- Python (Pandas, NumPy , Matplotlib , Seaborn)
- IQR Outlier Detection
- Tableau Dashboard

BUSINESS IMPACT & STRATEGIC RECOMMENDATIONS

- **Inventory Prioritization**
  - Focus on Kitchen & Garden top SKUs
- **Geographic Targeting**
  - Invest in Australia and design premium offers for Hong Kong
- **Forecasting**
  - Implement demand forecasting models
- **Marketing & Retention**
  - Target high-value and frequent buyers with personalized offers.

FINAL INSIGHTS & ANALYTICAL PERSPECTIVE

❖ **After completing the analysis, the following strategic observations stand out:**

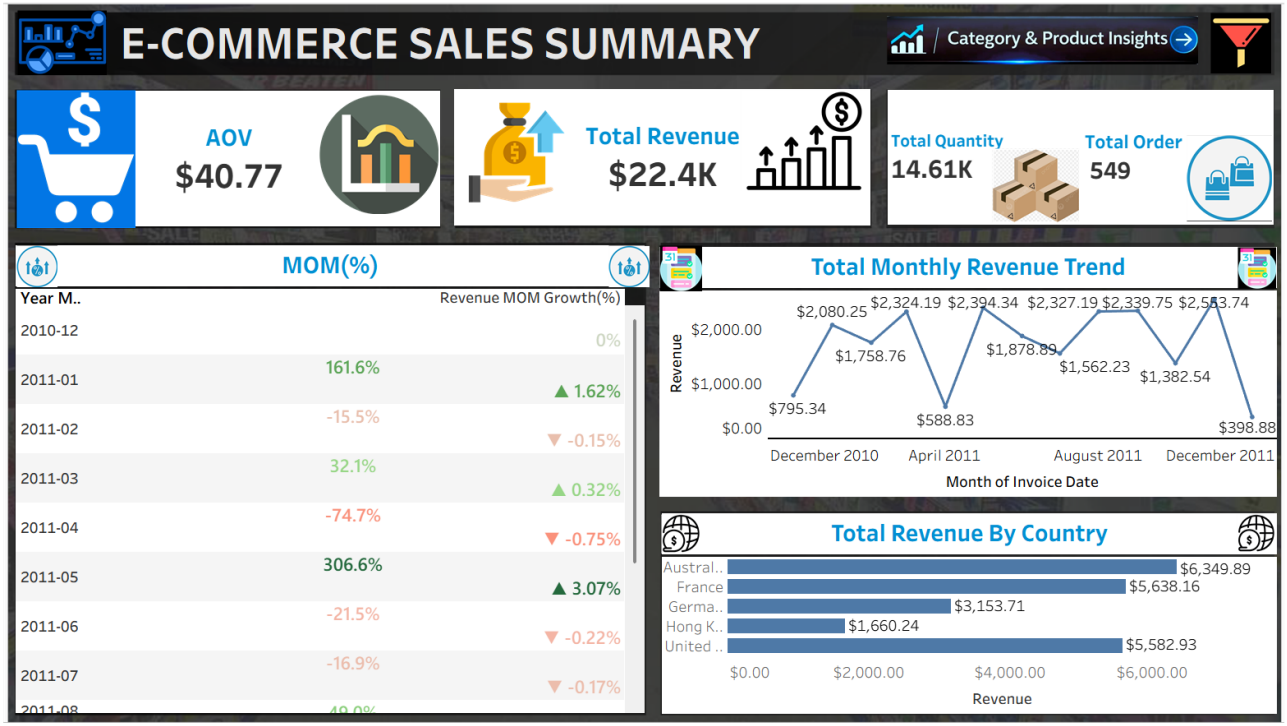
- Revenue is highly concentrated in a limited number of categories, especially Kitchen & Garden, indicating dependency risk but strong core demand.
- Australia generates the highest total revenue (~$6.3K), making it a priority market for expansion and targeted marketing campaigns
- Hong Kong shows exceptionally high AOV (~$237), suggesting a premium customer segment that can be leveraged for high-margin products.
- Seasonal revenue fluctuations indicate demand cycles, which can be utilized for inventory optimization and promotional timing.
- Customer revenue distribution suggests that a small segment contributes disproportionately to total revenue — indicating strong potential for loyalty programs and retention strategies.

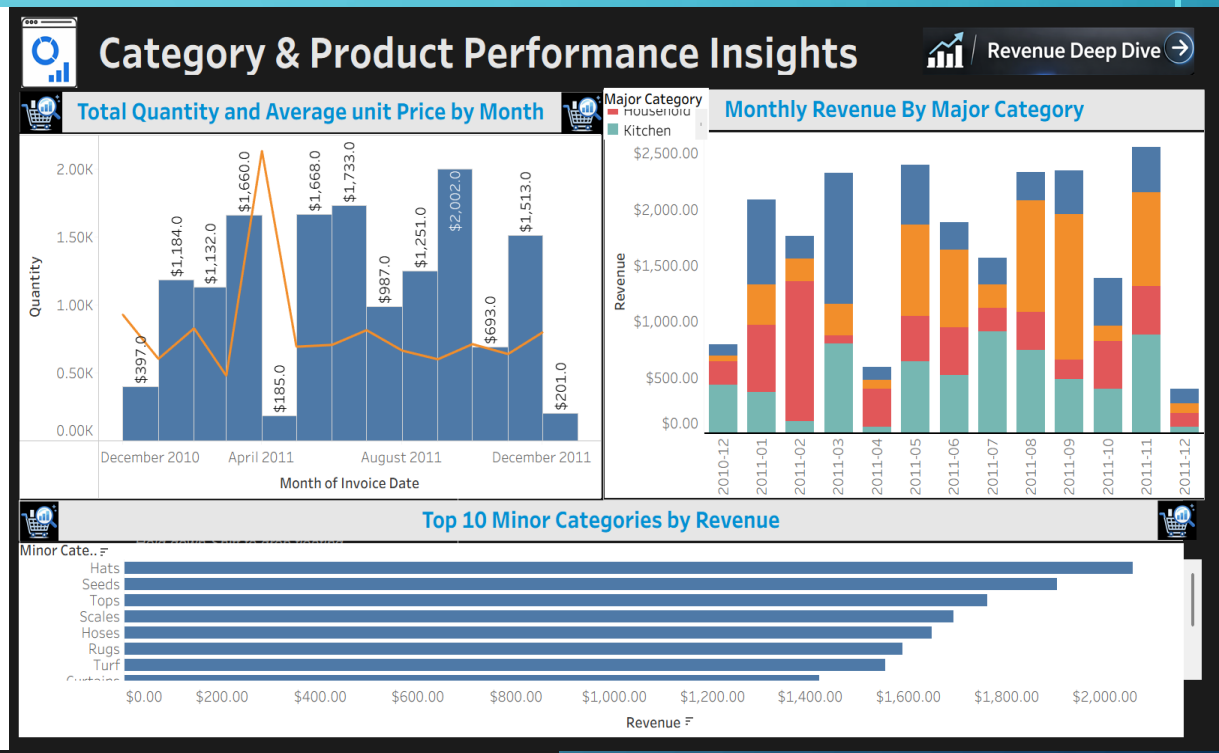
MY OPINION AS A DATA ANALYST

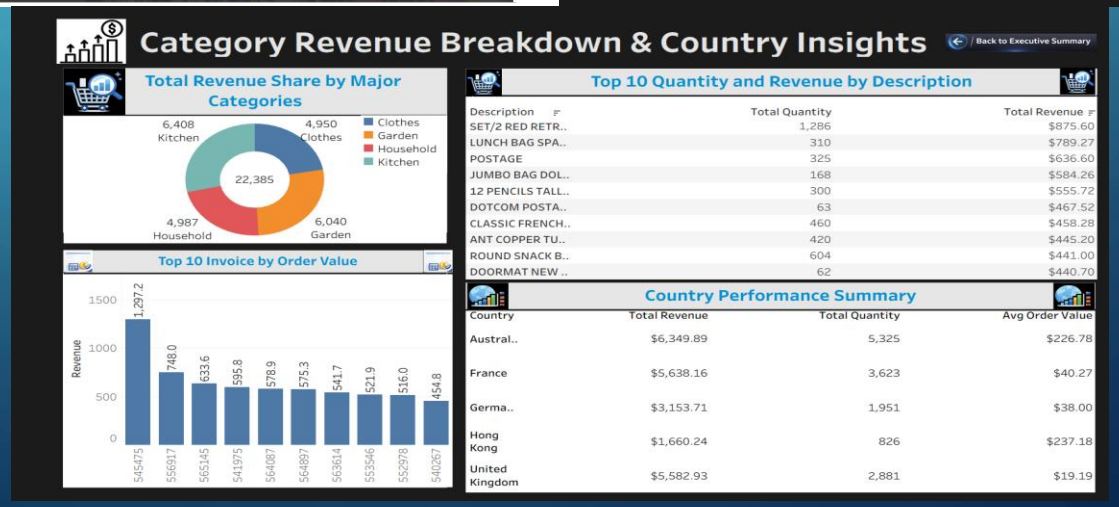
❖ **Based on the analysis the business should:**

- Focus on retaining high-value customers rather than only acquiring new ones.
- Strengthen premium positioning in high-AOV markets.
- Reduce over-dependence on a single dominant category by diversifying product performance.
- Use seasonal patterns to improve marketing ROI and inventory planning.

INTERACTIVE DASHBOARD







CONCLUSION

This project analyzed customer purchasing behavior to understand revenue patterns and key performance drivers. The findings highlight important categories, markets, and customer segments that contribute most to revenue These insights can support better business decisions in marketing, inventory planning, and overall growth strategy.

THANK YOU

Project Resources and Links:-

Dashboard / Project Repository Link:-